



Samsung AI scaling

All the details on the filters and AI technology used by Samsung to make videos look good on 8K displays. <https://dgit.in/apr20-71>



Impossible zoom in CSI

The famous scene where detectives identify a suspect by zooming into the reflection on the victim's iris. <https://dgit.in/apr20-72>

two or more competing algorithms that fight with each other to produce the better image. The upscaled image is progressively pushed towards a more photorealistic output through the competition.

All of this research is great, it tells us that technology combined with artificial intelligence can improve the resolution of images. However, how does the end user get from it? Well, the cutting edge tech has actually found its way to many of the products that you actually use.

BENEFITS TO CONSUMERS

On the web, Google reduces the data consumed while retrieving images by using RAISR to pull a low resolution version of an image before restoring detail. Twitter uses the technology from its acquisition, Magic Pony, to upscale low quality videos.

The Pixel series of phones have consistently been giving consumers some of the most advanced AI developed by Google. For example, from the Pixel 3 onwards, Google has been implementing the technology that was originally developed to upscale images in innovative new ways. Instead of upscaling the image of a single frame, the image information from multiple frames are processed together to give a single image. Then, the technology was implemented in a feature called Super Res Zoom, which means that if you zoom in through the camera app and capture an image, you get a better result than simply capturing a single image and then cropping it to the required zoom level. The resulting digital zoom is comparable to the quality of 2x optical zoom in other smartphone cameras. To achieve this, Super Res Zoom is combined with RAISR. The same technology also helps in getting clear, noiseless images for the famous low light shots of the Pixel series.

If you see some of the old 360p or even 240p phone videos on the 4K

displays of today, they look horrible pixelated. Similarly, in a few years, the 720p and 1080p videos of today are bound to look horrible on 8K displays. Even when playing content from Blue Ray discs, 4K TVs need to upscale by 4 times, and 8K by 8 times. To achieve this, Samsung developed over 100 filters by training their AI on various kinds of content. So depending on whether you are watching a football match, an F1 race, or an action movie... the AI knows what kind of upscaling technique to apply. This requires balancing various factors such as detail creation, noise reduction and edge restoration, to provide the best video in the end. The technology is available in its 8K televisions. Other television manufacturers also use technology on similar lines. LG uses something it calls ThinQ AI, which upscales content for its 8K televisions. The technology is also trained on over a million pieces of content. Sharpness and vividity are

The NVIDIA Shield TV is a streaming media player with the capability of upscaling 720p videos to 4K. The tech works in realtime, with the content from streaming service providers. AI-based upscaling is also embedded into its NGX SDK, which is used by a number of developers for creating their applications. A startup called Entropix uses NVIDIA Tesla V100 GPUs in tandem with TensorRT to upscale videos for a number of potential applications, that can give better imaging capabilities to autonomous vehicles, or more quickly identify lost children in a crowd. AMD does something interesting with Virtual Super Resolution, where the games actually render the content in 4K, even if the user does not have a 4K display. The resulting image is then downsampled to the resolution of the monitor, allowing lower resolution monitors to approximate the image quality of 4K displays. The approach also allows AMD to bypass engine



The Up-Res capability of the NVIDIA NGX SDK

conserved or enhanced, while every effort is made to reduce noise. Sony uses technology it calls 4K X-Reality PRO, which are a set of algorithms to upscale images in its 4K TVs. In fact, there is a dedicated processor for the task, called the 4K X-Reality PRO. The patterns in the images are compared with a database, to decide the HSB (hue, saturation, brightness) levels for each pixel, a process that reduces noise in the upscaled images. Sony claims that it uses the expertise gained from producing and filming movies, towards the upscaling technology used in its consumer televisions.

and game implementations of other technologies it uses, providing a high quality experience to the end user.

In case you want to get to upscaling particular images yourself, the best option is a batch processing software called Gigapixel AI by Topaz labs. It is an expensive piece of software, costing \$99, but it is a one time purchase with no subscription fees. The AI used has been trained on thousands of low resolution images paired with high resolution ones. Images can be upscaled by up to 600 percent without any visible loss in quality or detail. There is also a video version available. [\[1\]](#)